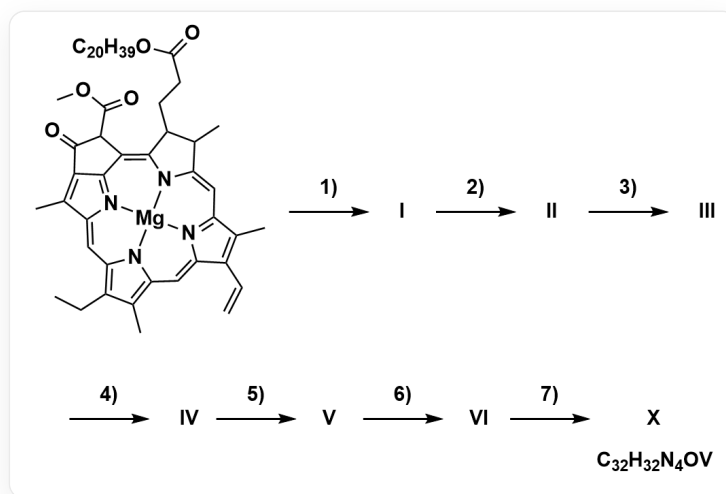


Question

Vanadium complexes can be synthesized using the following process:



SMILES structure of the substrate:

O=C(CCC1C(/C2=C/C3=N/C(C(C=C)=C3C)=C\C4=C(C(C(C)=C5N4[Mg]N2/C1=C(C(C6=C/7C)=NC7=C/5)/C(C6=O)C(OC)=O)C)C)O[R]

where R is $\text{C}_{20}\text{H}_{39}$, undergoes seven steps of reaction 1)–7) (described in subsequent questions), generating intermediate products

I/II/III/IV/V/VI and the final product X, with the chemical formula $\text{C}_{32}\text{H}_{32}\text{N}_4\text{OV}$

The following table lists the seven steps of the reaction and their reaction types:

Step	Reaction Type
1)	Metal Coordination Dissociation
2)	Hydrolysis
3)	Demethoxycarbonylation
4)	Reduction (gaining 4 electrons)
5)	Aromatization
6)	Decarboxylation
7)	Metal Chelation Coordination

For the unknown intermediate products **I** ~ **VI**, their chemical formulas can be written as:



Calculate the product of the mass fractions of each element, denoted as:

$$P_i = \prod_{j \in \{C, H, O, N\}} \omega_j, \quad i = \mathbf{I} \sim \mathbf{VI}$$

Based on the chemical formulas of the reactants and final product, deduce the chemical formulas of the intermediate products, calculate their P_i values, and finally, calculate the sum of all P_i :

$$S = \sum_i P_i, \quad i = \mathbf{I} \sim \mathbf{VI}$$

Choose the correct result from the following options. (Keep three significant figures for the calculation result, and choose the option with a deviation of less than 0.5% from the calculated result, otherwise choose **A**: None of the other options are correct)

A. All other options are incorrect

B. 0.00288

C. 0.00207

D. 0.00113

E. 0.000942

F. 0.00235

G. 0.00185

H. 0.00208

I. 0.00301

J. 0.00262

K. 0.00129

L. 0.00101

Answer

Correct Answer: **C**

Detailed Explanation

To solve this problem, we need to analyze the entire reaction process, determine the chemical formula of each intermediate, and then calculate the sum of the products of the mass fractions as required by the question.

First, we need to deduce the exact chemical formulas of intermediates **I** to **VI** based on the given seven-step reaction. The starting material for the reaction is a chlorophyll derivative, and according to the image information, its chemical formula is $C_{55}H_{72}N_4O_5Mg$.

CHECKPOINT

1 PTS

The chemical formula of the starting material is $C_{55}H_{72}N_4O_5Mg$

The first step is coordination metal dissociation, which means that the central magnesium ion is removed, usually replaced by two hydrogen atoms. Therefore, we generate intermediate **I** from the starting material, and its chemical formula is $C_{55}H_{74}N_4O_5$.

CHECKPOINT

1 PTS

The chemical formula of intermediate **I** is $C_{55}H_{74}N_4O_5$

The second step is hydrolysis, which usually acts on the ester group in the molecule. There are two ester groups in the molecule that can be hydrolyzed: one $C_{20}H_{39}$ ester group (commonly known as phytol group) and the other is a methyl ester group. If the methyl ester group is hydrolyzed in this step, the subsequent "demethoxycarbonyl" reaction cannot proceed.

CHECKPOINT

0.5 PTS

The methyl ester cannot be hydrolyzed in the second step

Therefore, only the phytol group can be hydrolyzed in this step. Hydrolysis will form a carboxylic acid and release phytol, resulting in the formation of intermediate **II**, and its chemical formula becomes $C_{35}H_{36}N_4O_5$.

CHECKPOINT

1 PTS

The chemical formula of intermediate **II** is $C_{35}H_{36}N_4O_5$

The third step is demethoxycarbonylation, which removes another smaller methoxycarbonyl group ($-COOCH_3$) from the molecule and replaces it with a hydrogen atom, resulting in a molecular formula of $C_{33}H_{34}N_4O_3$, which is intermediate **III**.

CHECKPOINT

1 PTS

The chemical formula of intermediate **III** is $C_{33}H_{34}N_4O_3$

The fourth step is a reduction reaction that gains 4 electrons. In porphyrin chemistry, this refers to the process of reducing a ketone group ($C=O$) on the ring to a methylene group (CH_2). This process consumes one oxygen atom and increases two hydrogen atoms, thus we obtain intermediate **IV**, and its chemical formula is $C_{33}H_{36}N_4O_2$.

CHECKPOINT

1 PTS

The chemical formula of intermediate **IV** is $C_{33}H_{36}N_4O_2$

The fifth step is aromatization. This process causes a partially saturated ring system to lose hydrogen atoms, forming a fully conjugated aromatic porphyrin macrocycle. Specifically, the molecule loses two hydrogen atoms, generating intermediate **V**, and its chemical formula is $C_{33}H_{34}N_4O_2$.

CHECKPOINT

1 PTS

The chemical formula of intermediate **V** is $C_{33}H_{34}N_4O_2$

The sixth step is decarboxylation, which removes the carboxyl group ($-COOH$) generated by the hydrolysis of the phytol group in the second step in the form of carbon dioxide (CO_2) and replaces it with a hydrogen atom, further changing the molecular formula to $C_{32}H_{34}N_4$, which is intermediate **VI**.

CHECKPOINT

1 PTS

The chemical formula of intermediate **VI** is $C_{32}H_{34}N_4$

Finally, intermediate **VI** coordinates with tetravalent VO^{2+} to obtain a vanadium complex with the chemical formula $C_{32}H_{32}N_4OV$.

Next, we need to calculate the P value of each intermediate, which is the product of the mass fractions of the four elements carbon, hydrogen, oxygen, and nitrogen it contains. We will use the following atomic weights for calculation: C: 12.011, H: 1.008, N: 14.007, O: 15.999.

For intermediate **I**, $C_{55}H_{74}N_4O_5$, its molar mass $M_I = 871.220$ g/mol. The corresponding product of mass fractions P_I is:

$$P_I = \left(\frac{55 \times 12.011}{871.220} \right) \times \left(\frac{74 \times 1.008}{871.220} \right) \times \left(\frac{4 \times 14.007}{871.220} \right) \times \left(\frac{5 \times 15.999}{871.220} \right) \approx 0.0003834$$

CHECKPOINT

1 PTS

$$P_I = 0.0003834$$

For intermediate **II**, $C_{35}H_{36}N_4O_5$, its molar mass $M_{II} = 592.696$ g/mol. The corresponding product of mass fractions P_{II} is:

$$P_{II} = \left(\frac{35 \times 12.011}{592.696} \right) \times \left(\frac{36 \times 1.008}{592.696} \right) \times \left(\frac{4 \times 14.007}{592.696} \right) \times \left(\frac{5 \times 15.999}{592.696} \right) \approx 0.0005541$$

CHECKPOINT

1 PTS

$$P_{II} = 0.0005541$$

For intermediate **III**, $C_{33}H_{34}N_4O_3$, its molar mass $M_{III} = 534.660$ g/mol. The corresponding product of mass fractions P_{III} is:

$$P_{\text{III}} = \left(\frac{33 \times 12.011}{534.660} \right) \times \left(\frac{34 \times 1.008}{534.660} \right) \times \left(\frac{4 \times 14.007}{534.660} \right) \times \left(\frac{3 \times 15.999}{534.660} \right) \approx 0.0004470$$

CHECKPOINT

1 PTS

$$P_{\text{III}} = 0.0004470$$

For intermediate **IV**, $\text{C}_{33}\text{H}_{36}\text{N}_4\text{O}_2$, its molar mass $M_{\text{IV}} = 520.677$ g/mol. The corresponding product of mass fractions P_{IV} is:

$$P_{\text{IV}} = \left(\frac{33 \times 12.011}{520.677} \right) \times \left(\frac{36 \times 1.008}{520.677} \right) \times \left(\frac{4 \times 14.007}{520.677} \right) \times \left(\frac{2 \times 15.999}{520.677} \right) \approx 0.0003508$$

CHECKPOINT

1 PTS

$$P_{\text{IV}} = 0.0003508$$

For intermediate **V**, $\text{C}_{33}\text{H}_{34}\text{N}_4\text{O}_2$, its molar mass $M_{\text{V}} = 518.661$ g/mol. The corresponding product of mass fractions P_{V} is:

$$P_{\text{V}} = \left(\frac{33 \times 12.011}{518.661} \right) \times \left(\frac{34 \times 1.008}{518.661} \right) \times \left(\frac{4 \times 14.007}{518.661} \right) \times \left(\frac{2 \times 15.999}{518.661} \right) \approx 0.0003365$$

CHECKPOINT

1 PTS

$$P_{\text{V}} = 0.0003365$$

For intermediate **VI**, $\text{C}_{32}\text{H}_{34}\text{N}_4$, since its molecule does not contain oxygen atoms, the mass fraction of its oxygen element ω_{O} is zero. Therefore, the product of mass fractions P_{VI} must also be zero.

$$P_{VI} = 0$$

CHECKPOINT

1 PTS

$$P_{VI} = 0$$

In the last step, we add up all the calculated P values to get the final sum S .

$$S = P_I + P_{II} + P_{III} + P_{IV} + P_V + P_{VI}$$

$$S \approx 0.00207$$

Therefore, after a complete calculation, the sum S of the P values of all intermediates is 0.00207.

CHECKPOINT

1 PTS

$$S = 0.00207$$